Module 3

Methods of Data Collection

CONTENTS

- ✓ Meaning and Types of data
- ✓ Methods of collecting primary data
- ✓ Difference between Questionnaire and Schedules
- ✓ Case study method
- ✓ Methods of collecting secondary data
- ✓ Selection of Appropriate Method for Data Collection
- ✓ Sampling: Types of sample design
- ✓ probability and non-probability sample techniques,
- ✓ Measurement and Scaling: Quantitative and qualitative data,
- ✓ Classification of measurement scales
- ✓ Scaling techniques – comparative and non-comparative scaling techniques

What Is Data Collection:

- **Data collection** is a process of
  - collecting information from all the relevant sources
  - to find answers to the research problem,
  - test the hypothesis

Data collection is the methodological process of gathering information about a specific subject.

Crucial to ensure your data is complete during the collection phase and that it's collected legally and ethically.

analysis won't be accurate and could have far-reaching consequences

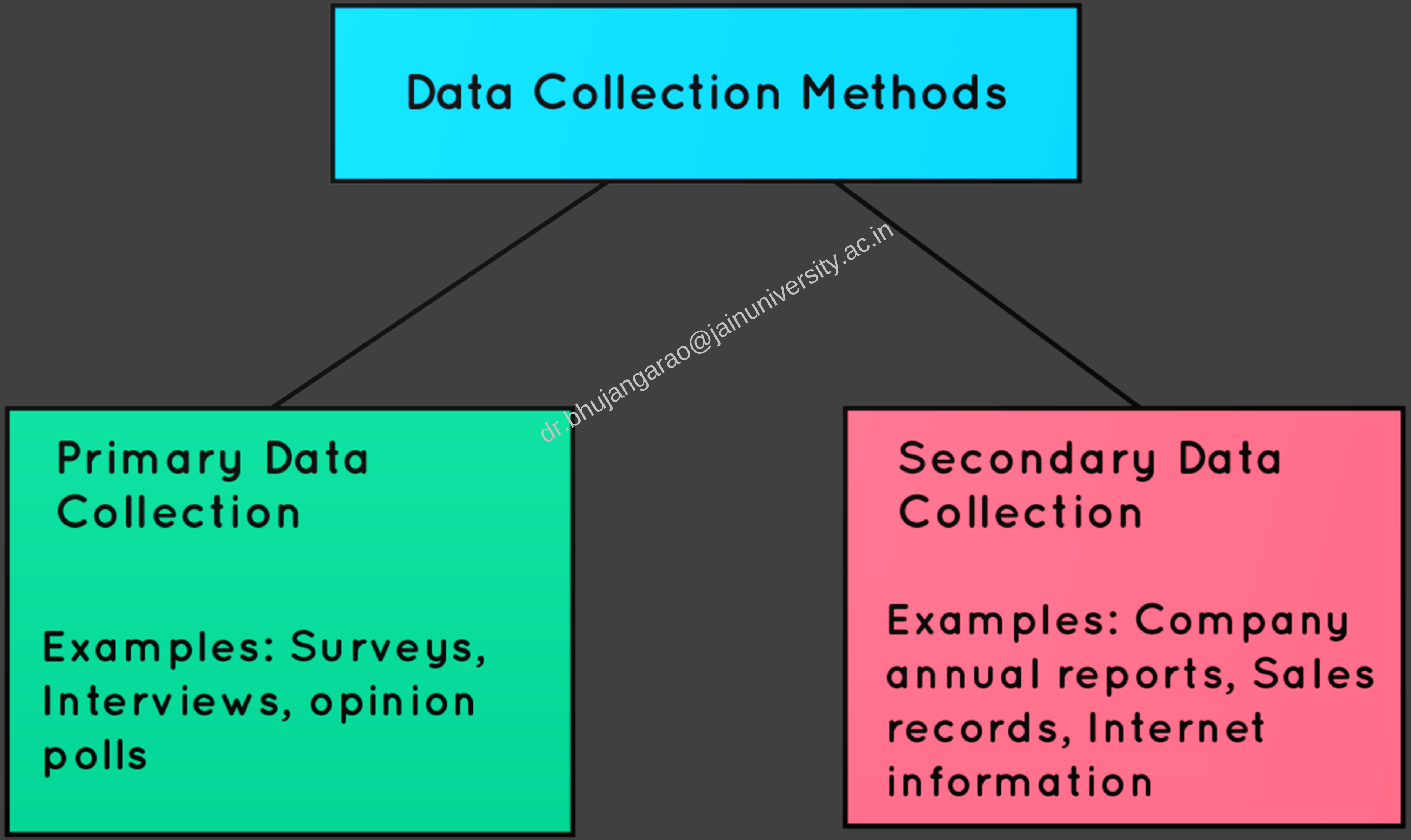
In general, there are three types of data:

First-party data, which is collected directly from users by your organization

Second-party data, which is data shared by another organization about its customers (or its first-party data)

Third-party data, which is data that's been aggregated and rented or sold by organizations that don't have a connection to your company or users

Types of Data Collection



DATA SOURCES

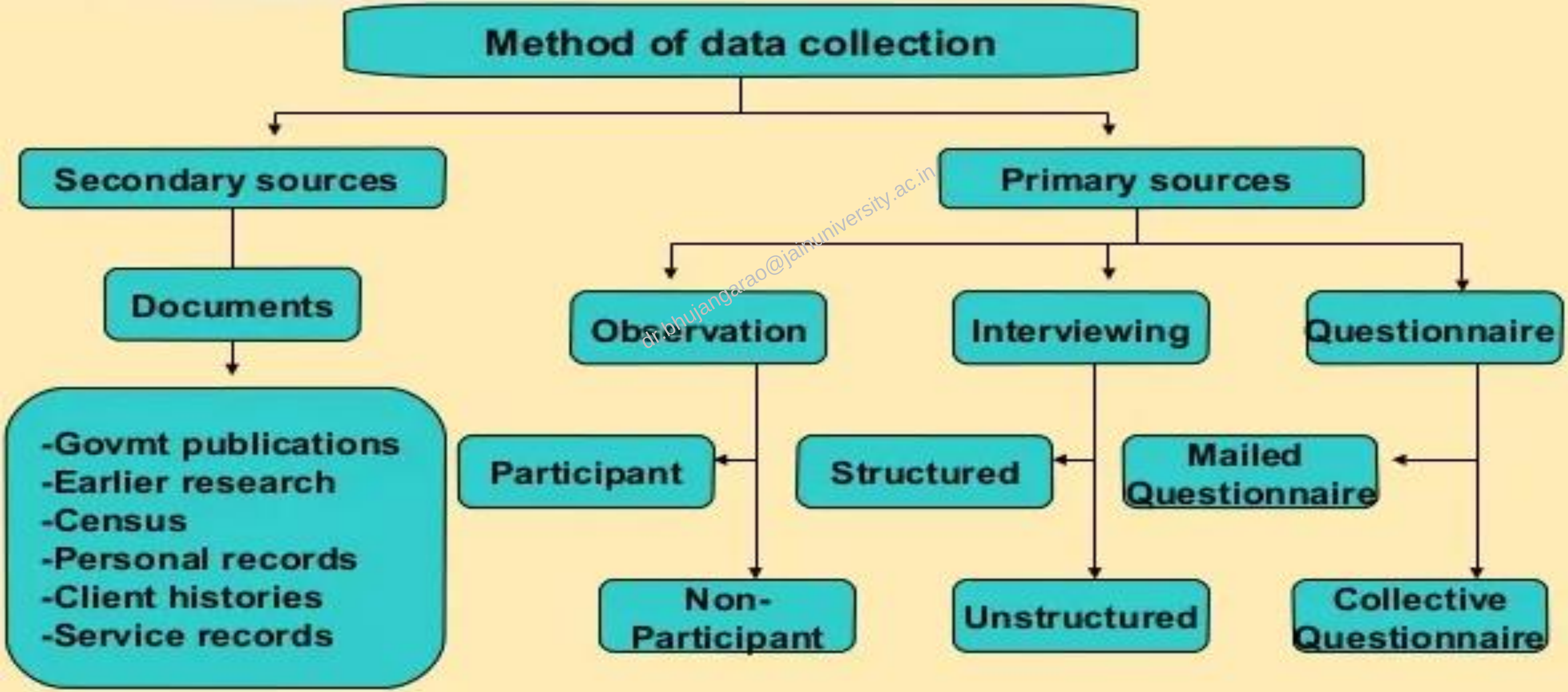
Primary Data

Primary data is data obtained directly from information sources directly. It means, a researcher or official which assigned have to come and hear the direct information from the respondent.

Secondary Data

Secondary data is data obtained not from sources of information directly. There is a lot of secondary data spread around the world.

Method of data collection



Collection of Primary Data

1. Observation
2. Interview
3. Questionnaire
4. Schedule
5. Content Analysis
6. Projective Techniques

Observation is a method of collecting primary data by observing directly and carefully a phenomenon, interaction, or issue that occurs systematically and structured.

Observation is **suitable** if we have a problem getting information from respondents who have difficulty in answering the question or information we want.

Advantages of observation:

– Researchers can get information from the sample directly

– Suitable for use in difficult conditions using questions as a data collection method

Disadvantages of observation:

– Observation data tend to be biased especially if the population sample is aware that they are being observed.

– The results of one observation can be different from other observations

– Sometimes, observations are often not completed due to technical matters such as respondents who do not cooperate, recording devices with problems, etc.

OBSERVATION

Participant Observation

Participant observation is a method of collecting data in which a researcher is actively involved in a research group that is the subject of observation, whether known or unknown by members of the group.

Non Participant Observation

Non participant observation is the opposite of participant observation. In this method, the researcher is not involved in the activity of the group being observed but only serves as a passive observation observer who hears, records, sees, and observes all the activities of the group.

Interview

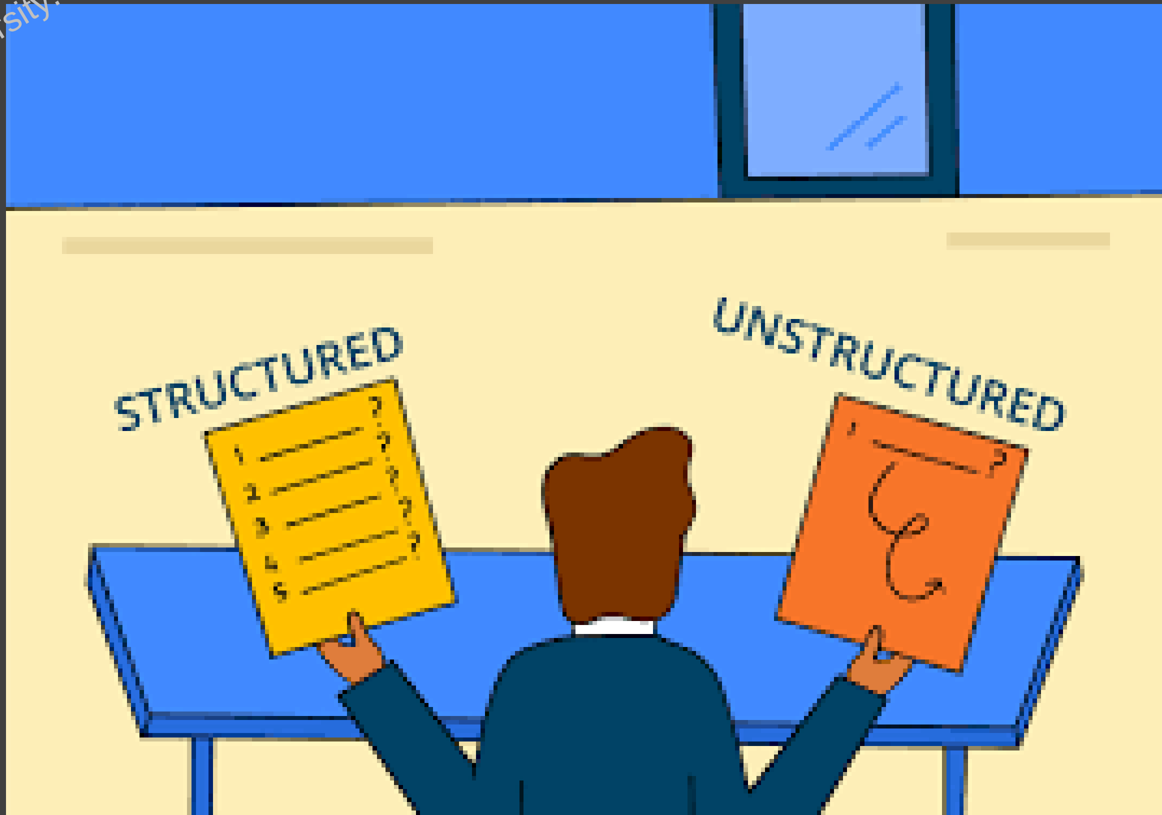
- ✓ most common method used in gathering information.
- ✓ interview is a question and answers interaction
- ✓ between two or more people with the aim to explore certain information.
- ✓ a researcher has the freedom to determine what will be asked to the respondent.
  - ➢ How to ask?
  - ➢ What words are used, and
  - ➢ How the order of questions.

You may use interviews in types of quantitative research or qualitative research.

It depends on the types of questions that you use.

There are 2 types of interviews:

1. Structured interview
2. Unstructured interviews



INTERVIEW

Structured interview

Structured interviews are interviews conducted with a number of rules, questions, and procedures that have been predetermined according to the implementation schedule.

Unstructured interviews

Unstructured interview is a type of question and answer that is free both in terms of context or structure. A researcher is free in determining the order and how to dig up that information.

3. Semi-Structured Interview

4. Personal Interview

5. Telephone Interview

6. Focus Group Interview

7. Depth Interview

8. Projective Techniques

3. Semi-Structured Interview

Additional questions might be asked during interviews to clarify and/or further expand certain issues.

4. Personal Interview

5. Telephone Interview

6. Focus Group Interview

Focus group interview is an unstructured interview which involves a moderator leading a discussion between a small groups of respondents on a specific topic.

7. Depth Interview

8. Projective Techniques

Interviews

ADVANTAGES

- Deep and free response
- Flexible, adaptable
- Glimpse into respondent's tone, gestures
- Ability to probe, follow-up, clarify misunderstandings about questions

DISADVANTAGES

- costly in time and personnel
- impractical with large numbers of respondents
- requires skill
- may be difficult to summarize responses
- possible biases: interviewer, respondent, situation

Questionnaire

A **questionnaire** is a research instrument that consists of a set of questions or other types of prompts that aims to collect information from a respondent.

Student Course Evaluation Questionnaire

(To be filled by each Student at the time of Course Completion)

Questionnaire	Strongly Agree	Agree	Uncertain	Disagree	Strongly Disagree
1. The course objectives were clear	☐	☐	☐	☐	☐
2. The Course workload was manageable	☐	☐	☐	☐	☐
3. The Course was well organized (e.g. timely access to materials, notification of changes, etc.)	☐	☐	☐	☐	☐
4. I think the Course was well structured to achieve the learning outcomes (there was a good balance of lectures, tutorials, practical etc.)	☐	☐	☐	☐	☐
5. The learning and teaching methods encouraged participation.	☐	☐	☐	☐	☐
6. The overall environment in the class was conducive to learning.	☐	☐	☐	☐	☐
7. Classrooms were satisfactory	☐	☐	☐	☐	☐
8. The Course stimulated my interest and thought on the subject area	☐	☐	☐	☐	☐
9. The pace of the Course was appropriate	☐	☐	☐	☐	☐
10. Ideas and concepts were presented clearly	☐	☐	☐	☐	☐
11. The method of assessment were reasonable	☐	☐	☐	☐	☐
12. Feedback on assessment was timely	☐	☐	☐	☐	☐
13. Feedback on assessment was helpful	☐	☐	☐	☐	☐
14. I understood the lectures	☐	☐	☐	☐	☐
15. The material was well organized and presented	☐	☐	☐	☐	☐
16. The instructor was responsive to student needs and problems	☐	☐	☐	☐	☐
17. Had the instructor been regular throughout the course?	☐	☐	☐	☐	☐
18. The material in the tutorials was useful	☐	☐	☐	☐	☐
19. I was happy with the amount of work needed for tutorials	☐	☐	☐	☐	☐
20. The tutor dealt effectively with my problems	☐	☐	☐	☐	☐

Ways of Administering a Questionnaire

The mailed questionnaire

Collective Administration

Administration in a public place

Advantages of Questionnaires

1. INEXPENSIVE
2. PRACTICAL
3. FAST RESULTS
4. SCALABILITY
5. COMPARABILITY
6. EASY ANALYSIS
7. VALIDITY & RELIABILITY
8. STANDARDIZED
9. NO PRESSURE
10. RESPONDENT ANONIMITY

Disadvantages of Questionnaires

1. DISHONEST ANSWERS
2. SKIPPED QUESTIONS
3. INTERPRETATION ISSUES
4. LACK OF NUANCE
5. ANALYSIS ISSUES
6. HIDDEN AGENDA
7. LACK OF PERSONALIZATION
8. UNCOSIENTIOUS RESPONSES
9. ACCESSIBILITY ISSUES
10. SURVEY FATIGUE

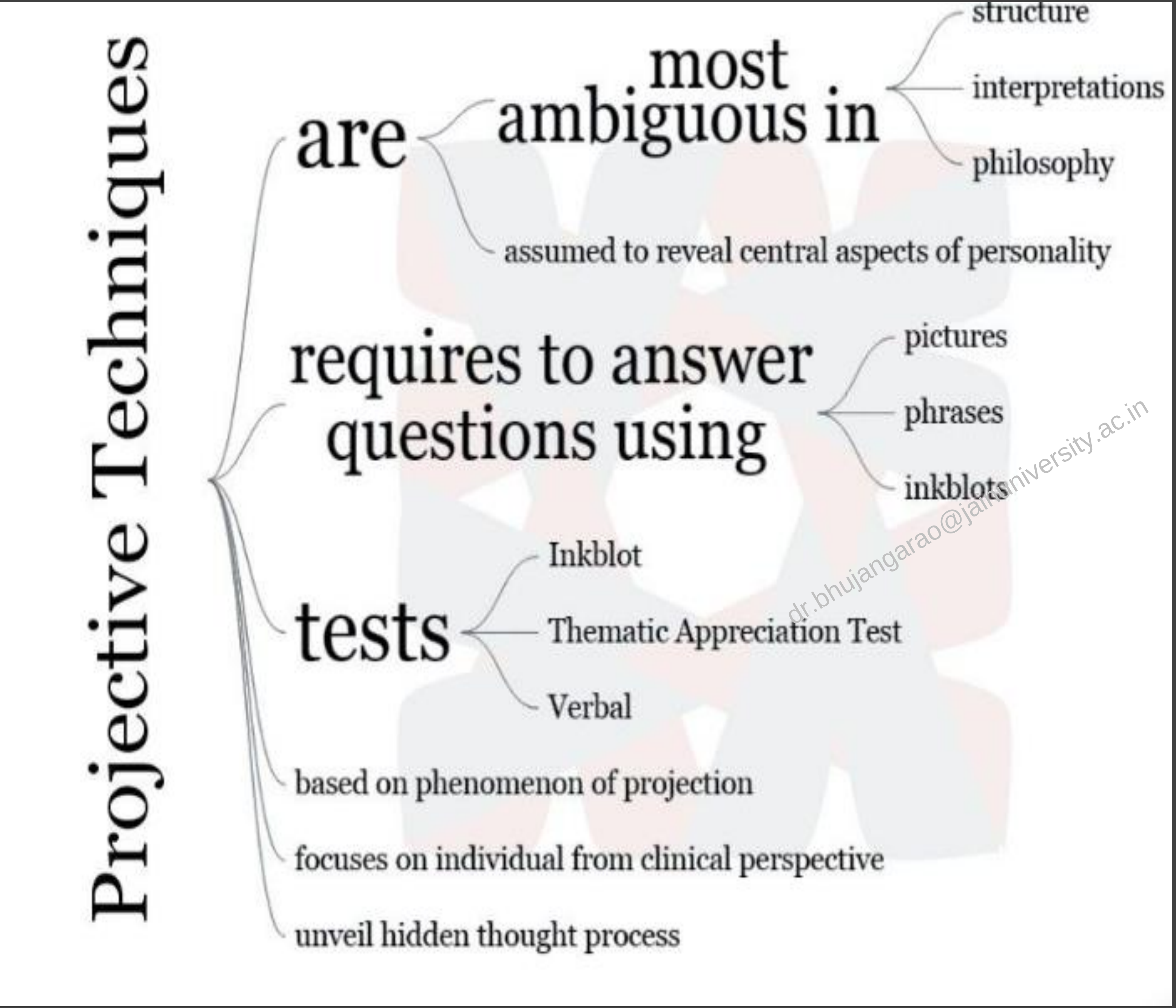
Schedule

Answers in the Schedule method of data collection are **filled by research workers/enumerators**.

Enumerator goes to the respondents, ask them the questions from the questionnaire in the order listed, and records the responses in the space provided.

Character	Questionnaire	Schedule
Delivered by	In general, questionnaires are delivered to the persons concerned either by post or mail, requesting them to answer the questions and return it.	Enumerators go to the informants with the schedule.
Role of Respondents	Read and understand the questions and reply in the space provided in the questionnaire itself.	Only answer the questions asked by enumerators. Sometimes, the schedule is distributed to the respondents, and the enumerators assist them in answering the questions.

Character	Questionnaire	Schedule
Filled by	Respondents	Enumerators
Response Rate	Low	High
Coverage	Large	Comparatively small
Cost	Economical	Expensive
Respondent's identity	Not known	Known
Observation Method	Not applicable	Applicable
rtant features required	· Simple to understand · Short questions · Interesting and Engaging	No special features required
Success relies on	Quality of the questionnaire	Honesty and competence of the enumerator.
Usage	Only when the people are literate and cooperative.	Used on both literate and illiterate people.


Projective Techniques
are
most ambiguous in
structure
interpretations
philosophy
assumed to reveal central aspects of personality
requires to answer questions using
pictures
phrases
inkblots
tests
Inkblot
Thematic Appreciation Test
Verbal
based on phenomenon of projection
focuses on individual from clinical perspective
unveil hidden thought process


Thanks for doing business with us today. Is there a way we can better serve you in the future?

Selection of Secondary Data

1. Reliability of Data
2. Suitability of Data
3. Adequacy of Data

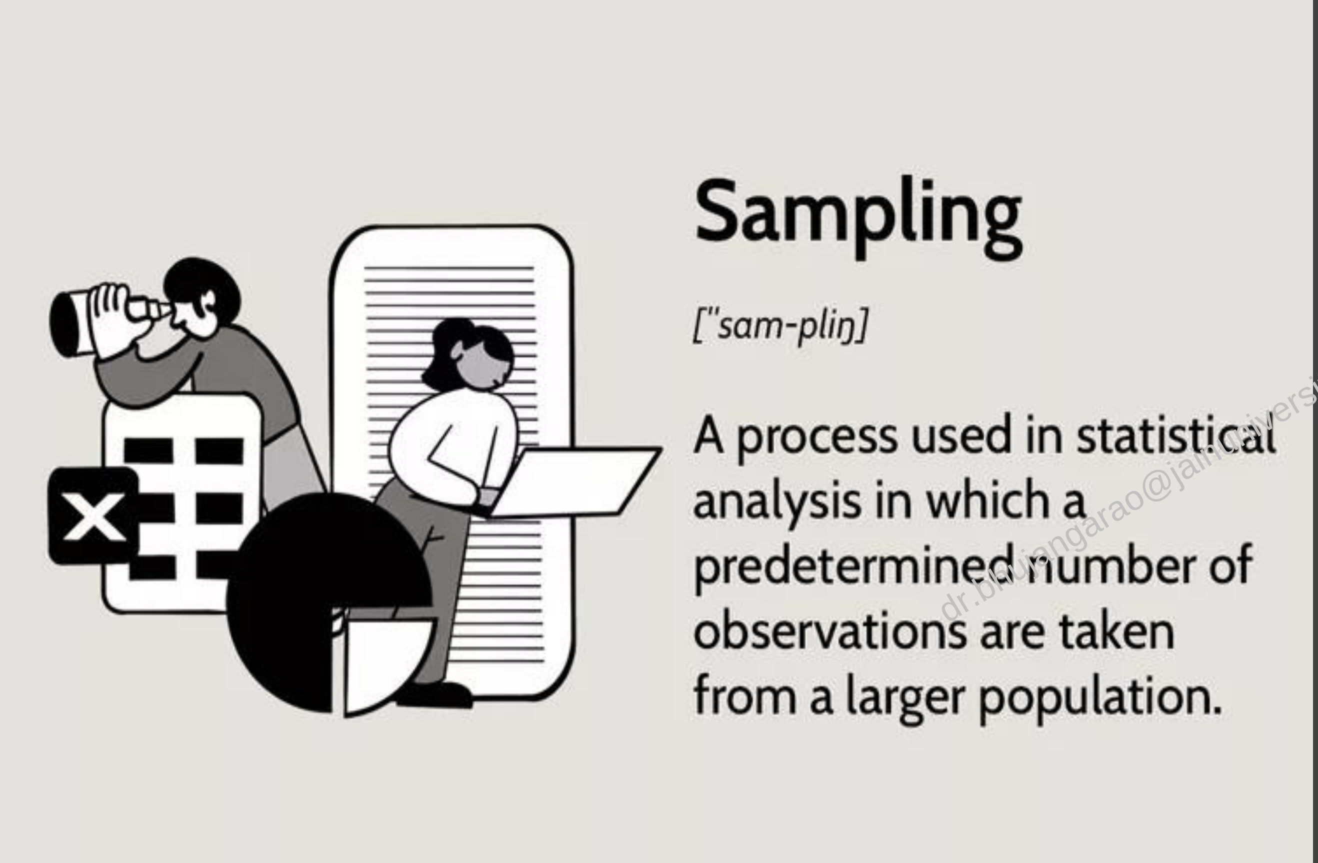
Collection of Secondary Data

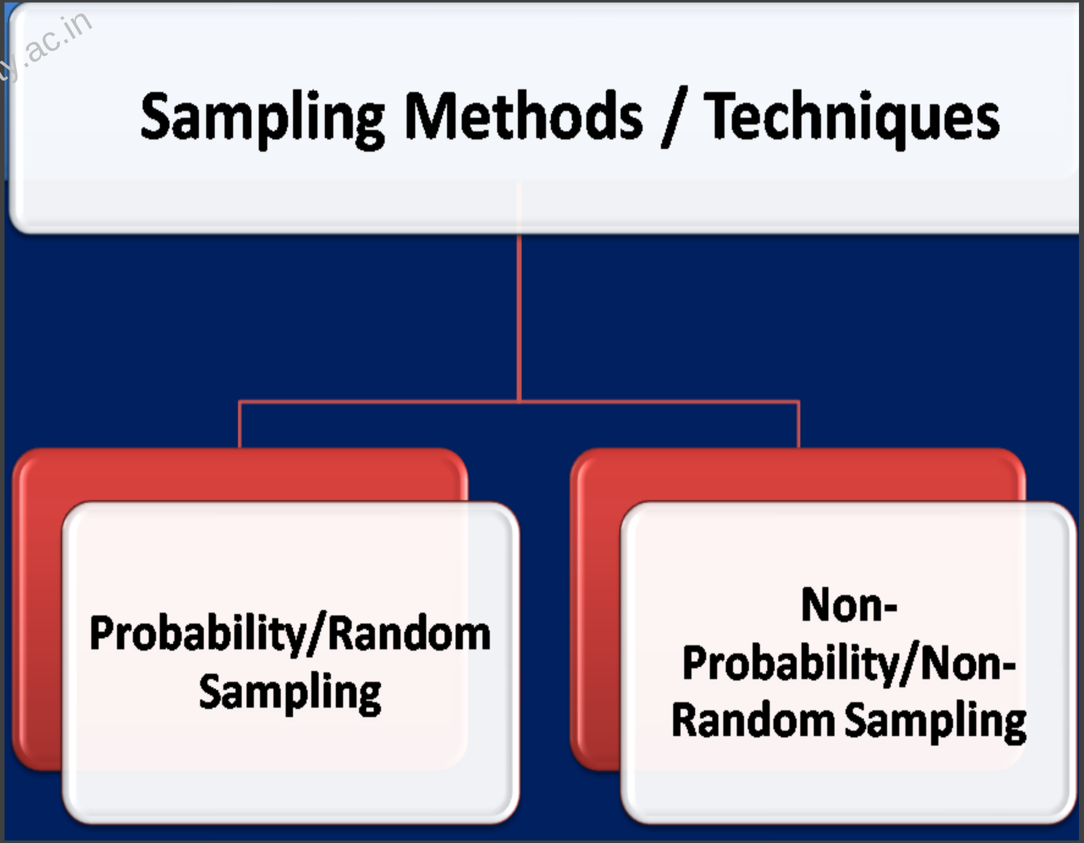
- **Identify the topic of research:** Before beginning secondary research, identify the topic that needs research. Once that's done, list down the research attributes and its purpose.
- **Identify research sources:** Next, narrow down on the information sources that will provide most relevant data and information applicable to your research.
- **Collect existing data:** Once the data collection sources are narrowed down, check for any previous data that is available which is closely related to the topic. Data related to research can be obtained from various sources like newspapers, public libraries, government and non-government agencies etc.
- **Combine and compare:** Once data is collected, combine and compare the data for any duplication and assemble data into a usable format. Make sure to collect data from authentic sources. Incorrect data can hamper research severely.
- **Analyze data:** Analyze data that is collected and identify if all questions are answered. If not, repeat the process if there is a need to dwell further into actionable insights.

Selection of Appropriate Method for Data Collection

1. Purpose and scope of enquiry.
2. Availability of time.
3. Availability of finance.
4. Accuracy required.
5. Statistical tools to be used.
6. Sources of information (data).
7. Method of Data Collection

SAMPLING TECHNIQUES





Probability Sampling Methods

Probability sampling means that every member of the population has a chance of being selected.

Non-Probability Sampling Methods

Non-probability sampling is a sampling method in which not all members of the population have an equal chance of participating in the study,

Probability Sampling Methods

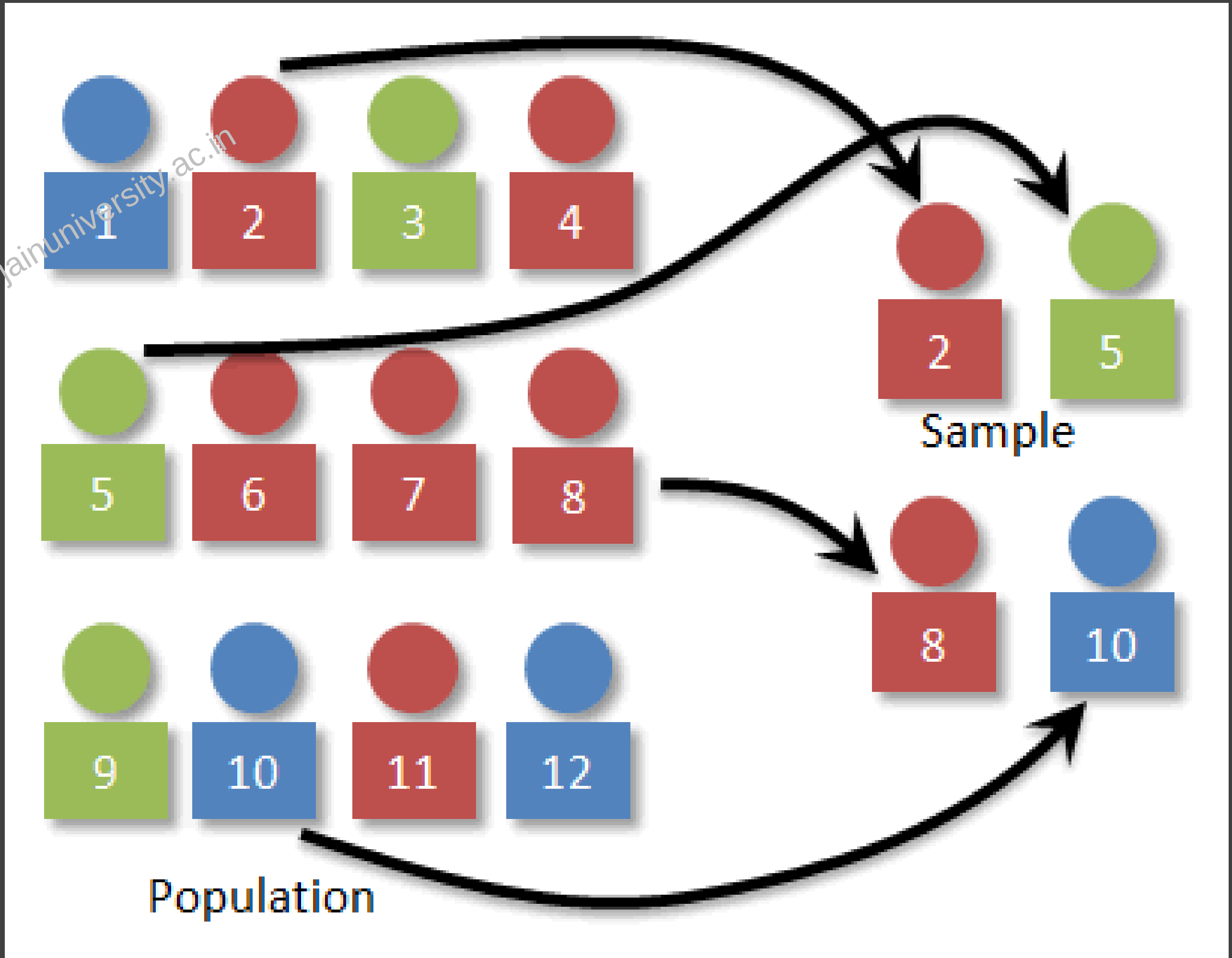
1. Simple random sampling
2. Systematic sampling
3. Stratified sampling
4. Clustered sampling

Non-Probability Sampling Methods

1. Convenience sampling
2. Quota sampling
3. Judgement (or Purposive) Sampling
4. Snowball sampling

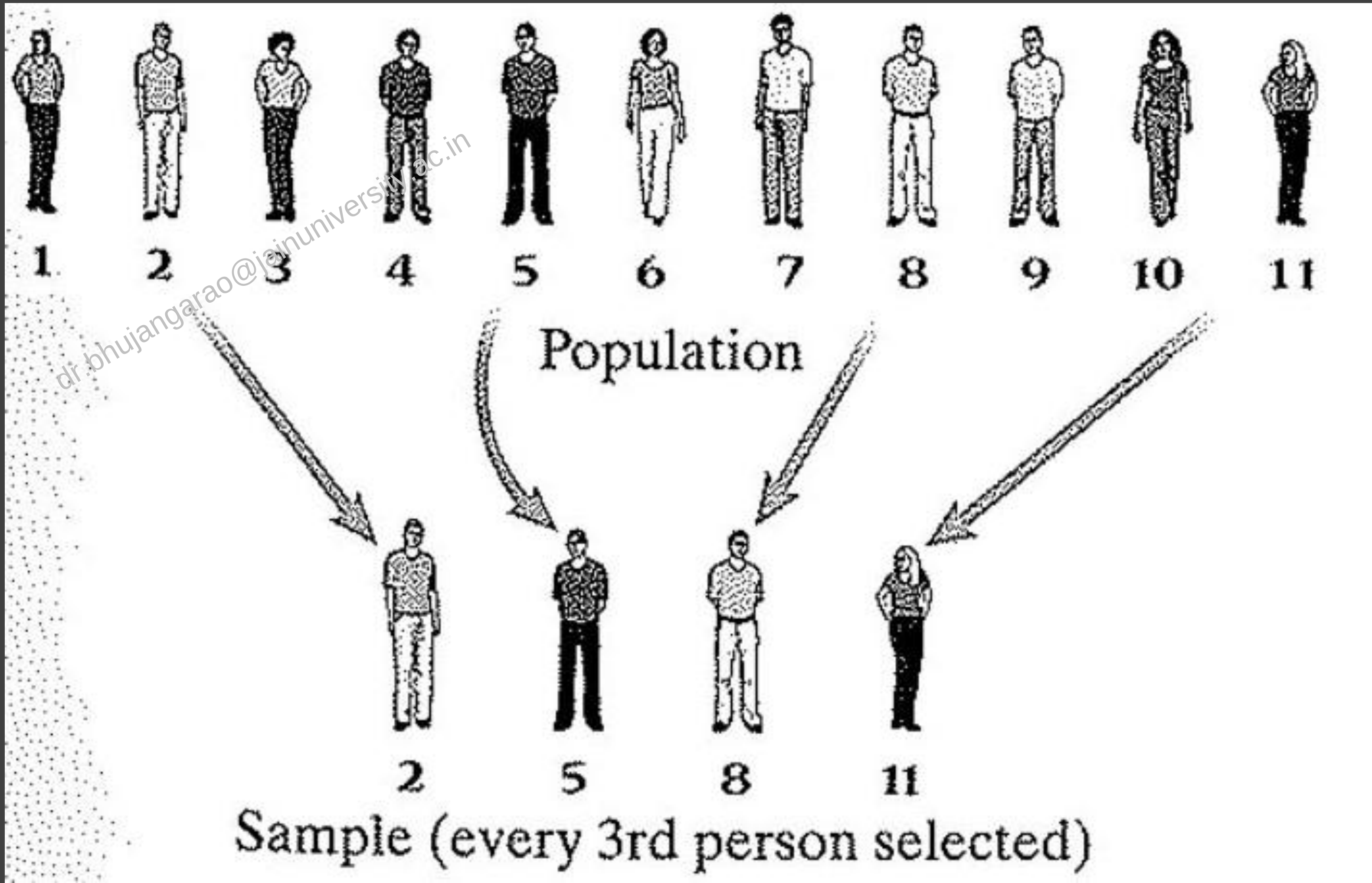
1. Simple random sampling

very member of the population has an equal chance of being selected.



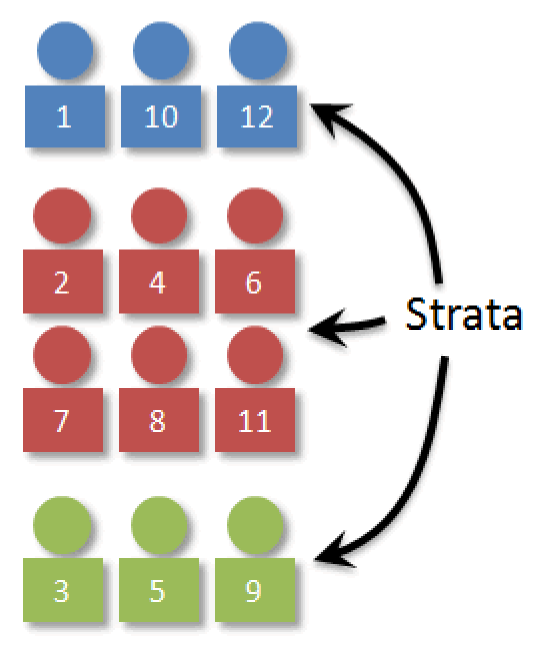
2. Systematic sampling

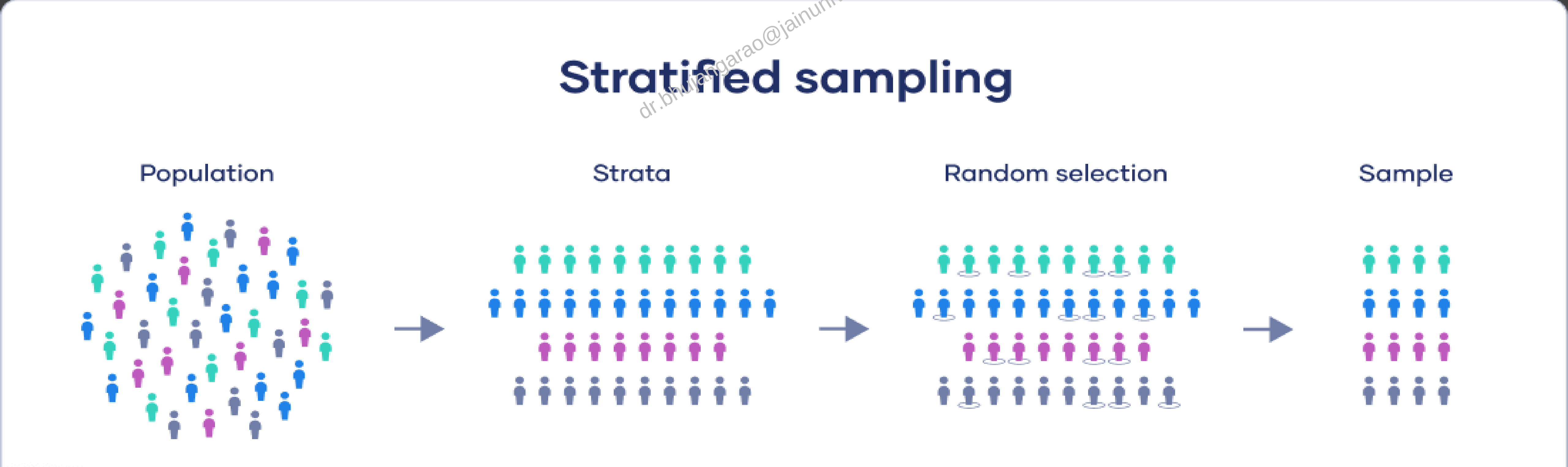
Every member of the population is listed with a number,



3. Stratified sampling

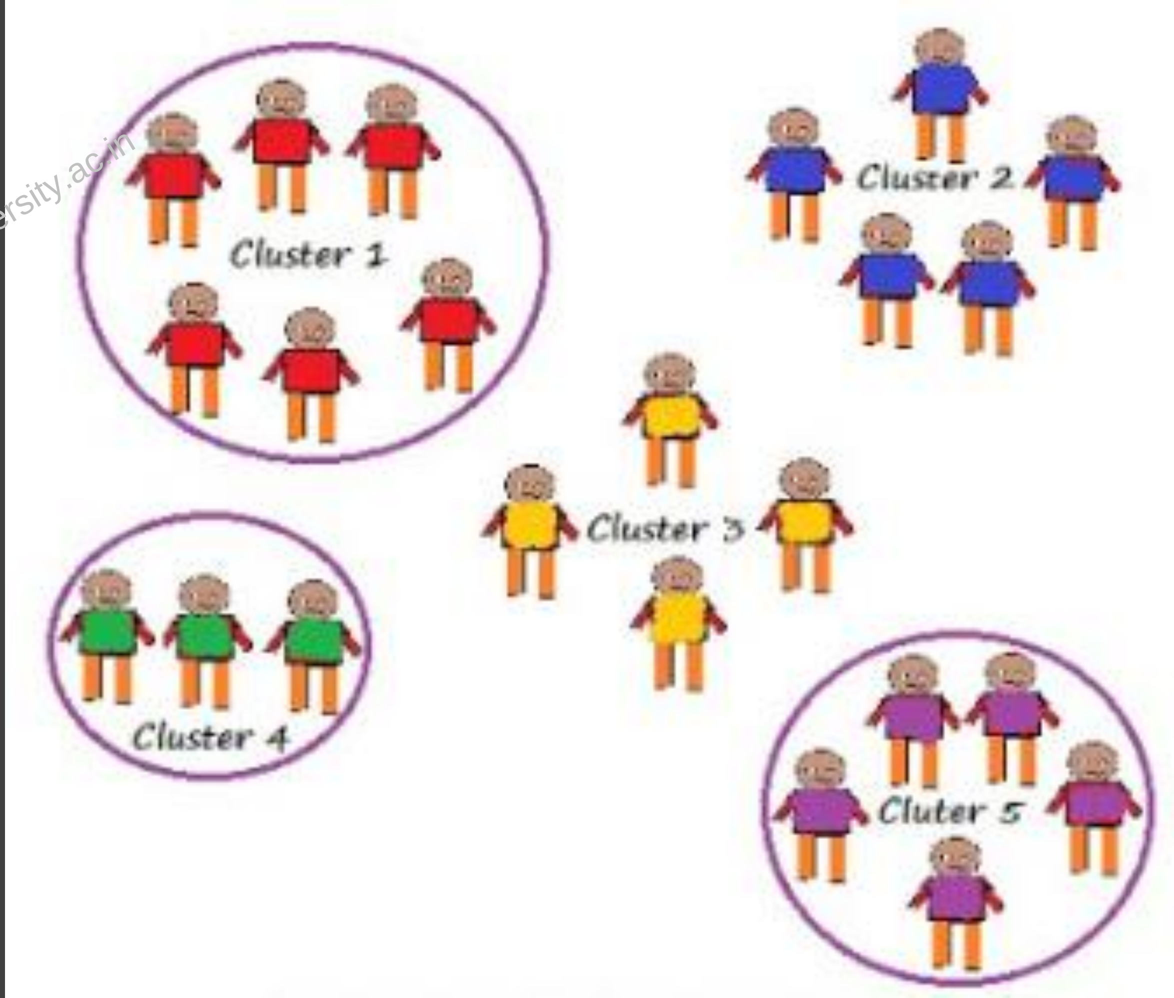
dividing the population into subpopulations (called strata)





4. Cluster sampling

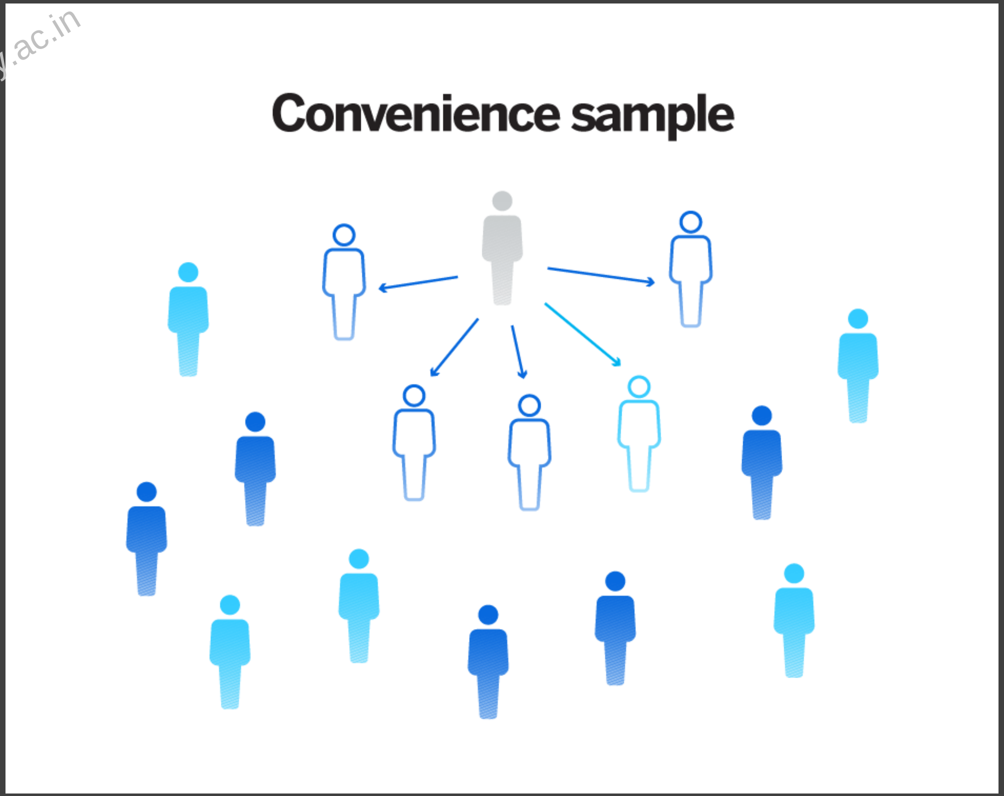
dividing the population into subgroups



Non-Probability Sampling Methods

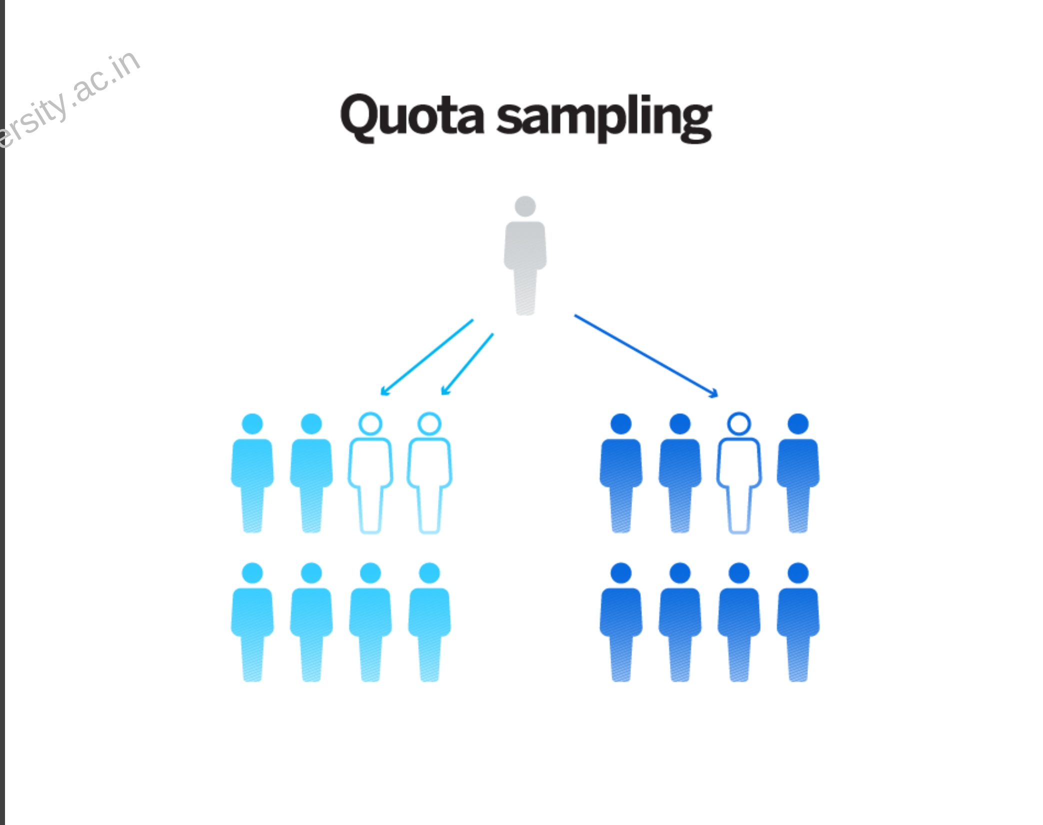
Convenience sampling

determined by convenience to the researcher.



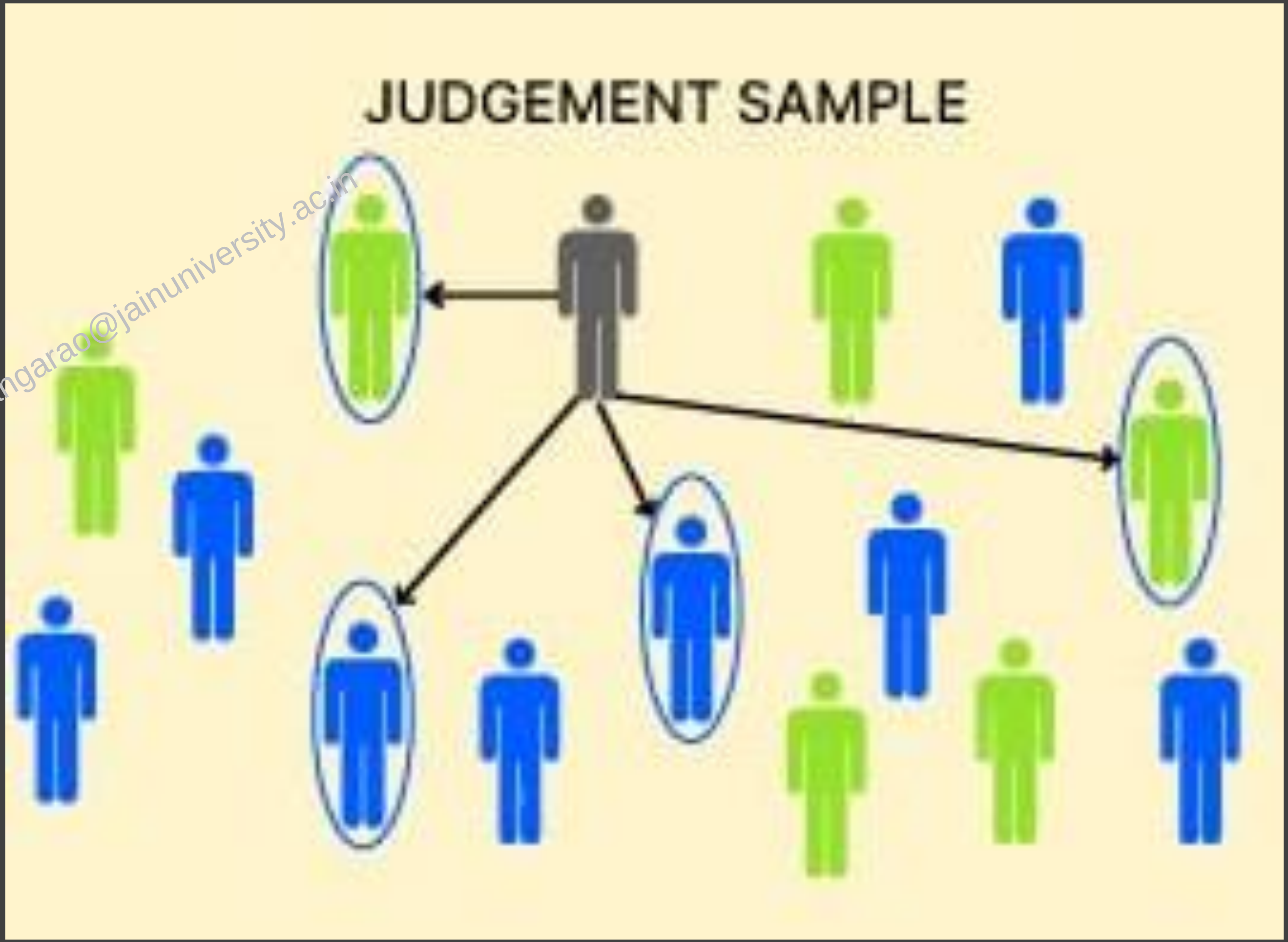
Quota sampling

predetermined number or proportion of units



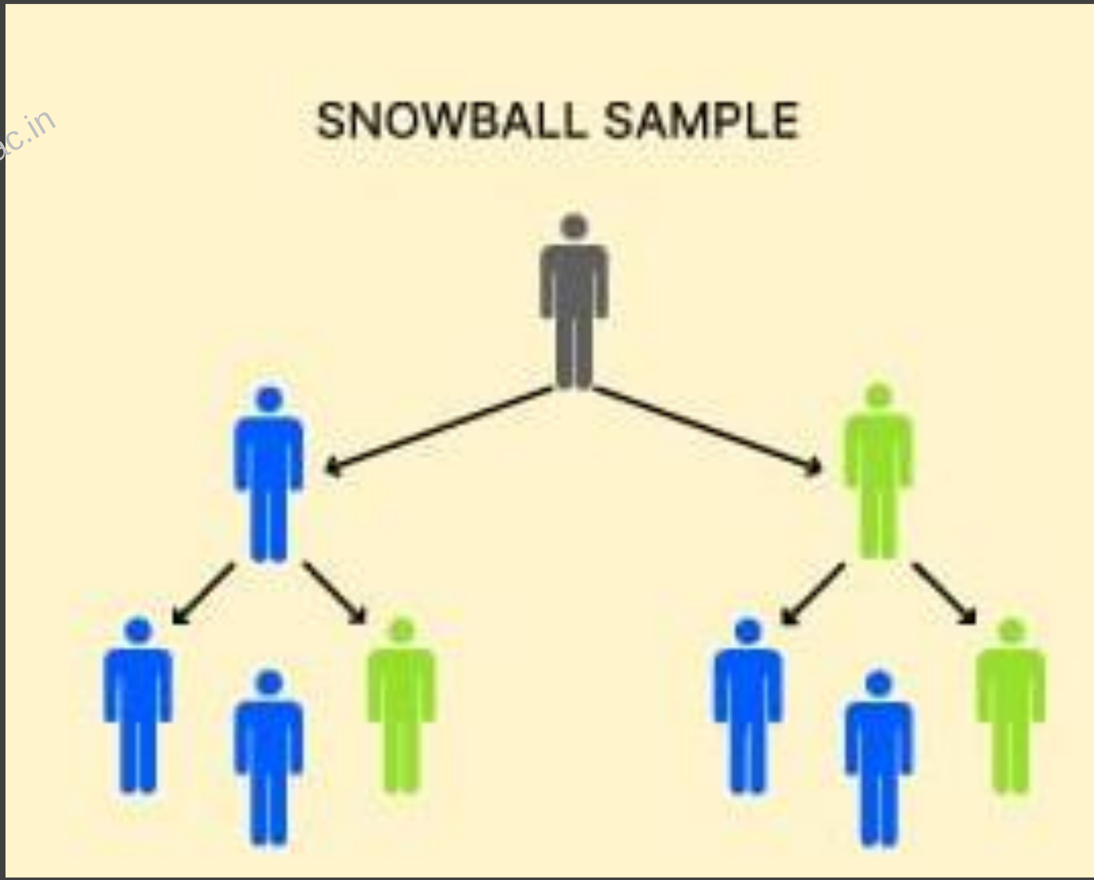
Purpose (judgmental) sampling

choose participants deliberately due to qualities they possess.



Snowball sampling

Start by finding one person who is willing to participate in your research. You then ask them to introduce you to others.



Scales of Measurement

In Statistics, the variables or numbers are defined and categorised using different **scales of measurements**. Each **level of measurement** scale has specific properties that determine the various use of statistical analysis. In this article, we will learn four types of scales such as nominal, ordinal, interval and ratio scale.

What is the Scale?

A scale is a device or an object used to measure or quantify any event or another object.

Levels of Measurements

There are four different scales of measurement. The data can be defined as being one of the four scales. The four types of scales are:

- Nominal Scale
- Ordinal Scale
- Interval Scale
- Ratio Scale

Nominal Scale

A nominal scale is the 1st level of measurement scale in which the numbers serve as “tags” or “labels” to classify or identify the objects. A nominal scale usually deals with the non-numeric variables or the numbers that do not have any value.

Characteristics of Nominal Scale

- A nominal scale variable is classified into two or more categories. In this measurement mechanism, the answer should fall into either of the classes.

- It is qualitative. The numbers are used here to identify the objects.

- The numbers don’t define the object characteristics. The only permissible aspect of numbers in the nominal scale is “counting.”

Example:

An example of a nominal scale measurement is given below:

What is your gender?

M- Male
F- Female

Here, the variables are used as tags, and the answer to this question should be either M or F.

Ordinal Scale

The ordinal scale is the 2nd level of measurement that reports the ordering and ranking of data without establishing the degree of variation between them. Ordinal represents the “order.” Ordinal data is known as qualitative data or categorical data. It can be grouped, named and also ranked.

Characteristics of the Ordinal Scale

- The ordinal scale shows the relative ranking of the variables
- It identifies and describes the magnitude of a variable
- Along with the information provided by the nominal scale, ordinal scales give the rankings of those variables
- The interval properties are not known
- The surveyors can quickly analyse the degree of agreement concerning the identified order of variables

Example:

- Ranking of school students – 1st, 2nd, 3rd, etc.
- Ratings in restaurants
- Evaluating the frequency of occurrences
  - Very often
  - Often
  - Not often
  - Not at all
- Assessing the degree of agreement
  - Totally agree
  - Agree
  - Neutral
  - Disagree
  - Totally disagree

Interval Scale

The interval scale is the 3rd level of measurement scale. It is defined as a quantitative measurement scale in which the difference between the two variables is meaningful. In other words, the variables are measured in an exact manner, not as in a relative way in which the presence of zero is arbitrary.

Characteristics of Interval Scale:

- The interval scale is quantitative as it can quantify the difference between the values
- It allows calculating the mean and median of the variables
- To understand the difference between the variables, you can subtract the values between the variables
- The interval scale is the preferred scale in Statistics as it helps to assign any numerical values to arbitrary assessment such as feelings, calendar types, etc.

Example:

- Likert Scale
- Net Promoter Score (NPS)
- Bipolar Matrix Table

5-point scales

Satisfaction

1. Very dissatisfied
2. Dissatisfied
3. Neither dissatisfied or satisfied
4. Satisfied
5. Very satisfied

Likelihood

1. Very unlikely
2. Unlikely
3. Neutral
4. Likely
5. Very likely

Level of concern

1. Very unconcerned
2. Unconcerned
3. Neutral
4. Concerned
5. Very concerned

Agreement

1. Strongly disagree
2. Disagree
3. Neither agree or disagree
4. Agree
5. Strongly agree

Frequency

1. Never
2. Rarely
3. Sometimes
4. Often
5. Always

Awareness

1. Very unaware
2. Unaware
3. Neither aware or unaware
4. ware
5. Very aware

Familiarity

1. Very unfamiliar
2. Unfamiliar
3. Somewhat familiar
4. Familiar
5. Very familiar

Quality

1. Very poor
2. Poor
3. Acceptable
4. Good
5. Very good

Importance

1. Very unimportant
2. Unimportant
3. Neutral
4. Important
5. Very important

7-point scales

Agreement

1. Strongly disagree
2. Disagree
3. Somewhat disagree
4. Neither agree or disagree
5. Somewhat agree
6. Agree
7. Strongly agree

Frequency

1. Never
2. Rarely (less than 10% of the time)
3. Occasionally (about 30% of the time)
4. Sometimes (about 50% of the time)
5. Frequently (about 70% of time)
6. Usually (about 90% of the time)
7. Every time

Appropriateness

1. Absolutely inappropriate
2. Inappropriate
3. Slightly inappropriate
4. Neutral
5. Slightly appropriate
6. Appropriate
7. Absolutely appropriate

Satisfaction

1. Very dissatisfied
2. Dissatisfied
3. Slightly dissatisfied
4. Neutral
5. Slightly satisfied
6. Satisfied
7. Very satisfied

Reflective of me

1. Very untrue of me
2. Untrue of me
3. Somewhat untrue of me
4. Neutral
5. Somewhat true of me
6. True of me
7. Very true of me

Level of difficulty

1. Very easy
2. Easy
3. Somewhat easy
4. Neutral
5. Somewhat hard
6. Hard
7. Very hard

Priority

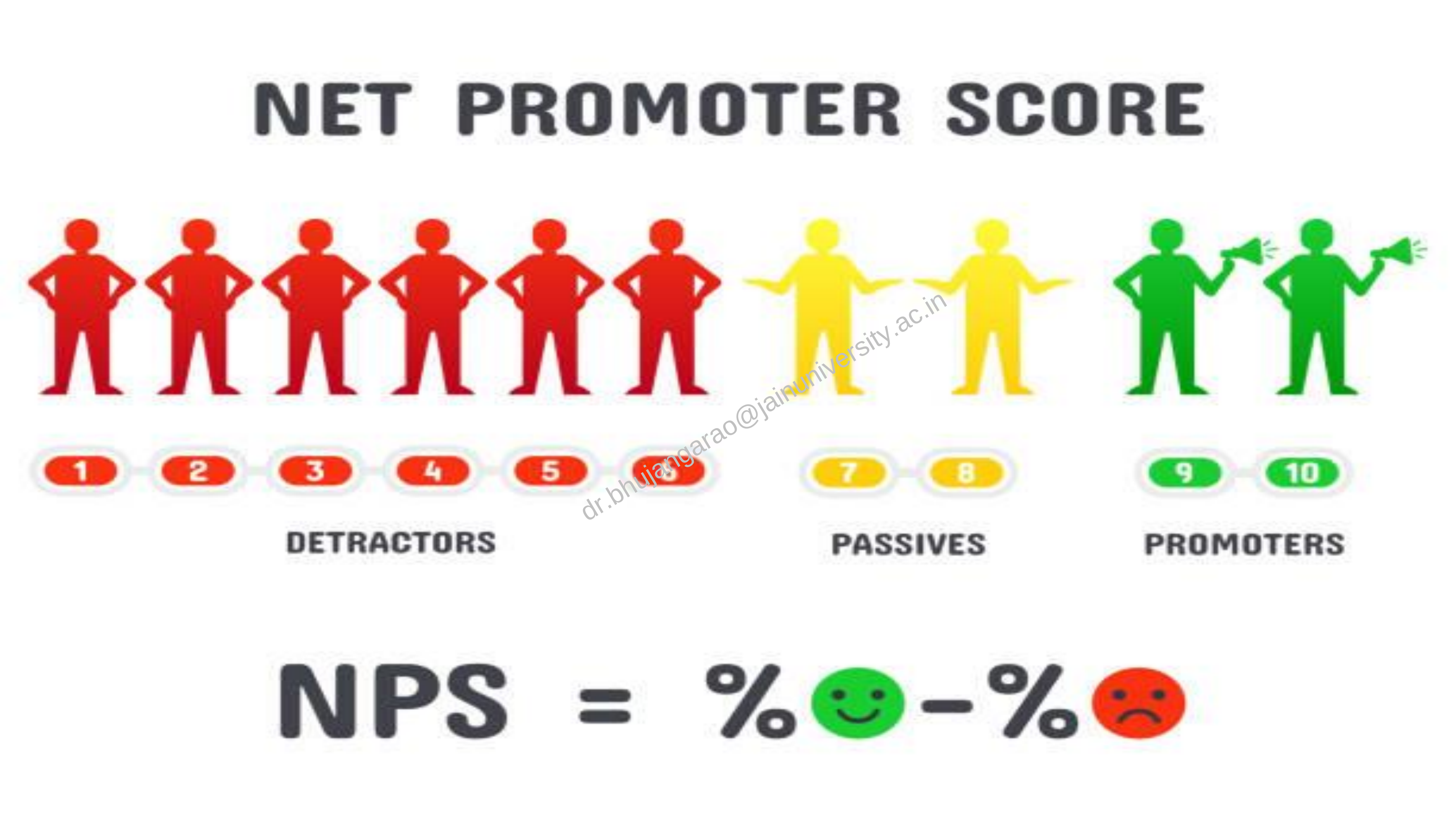
1. Not a priority
2. Low priority
3. Somewhat a priority
4. Neutral
5. Moderate priority
6. High priority
7. Essential priority

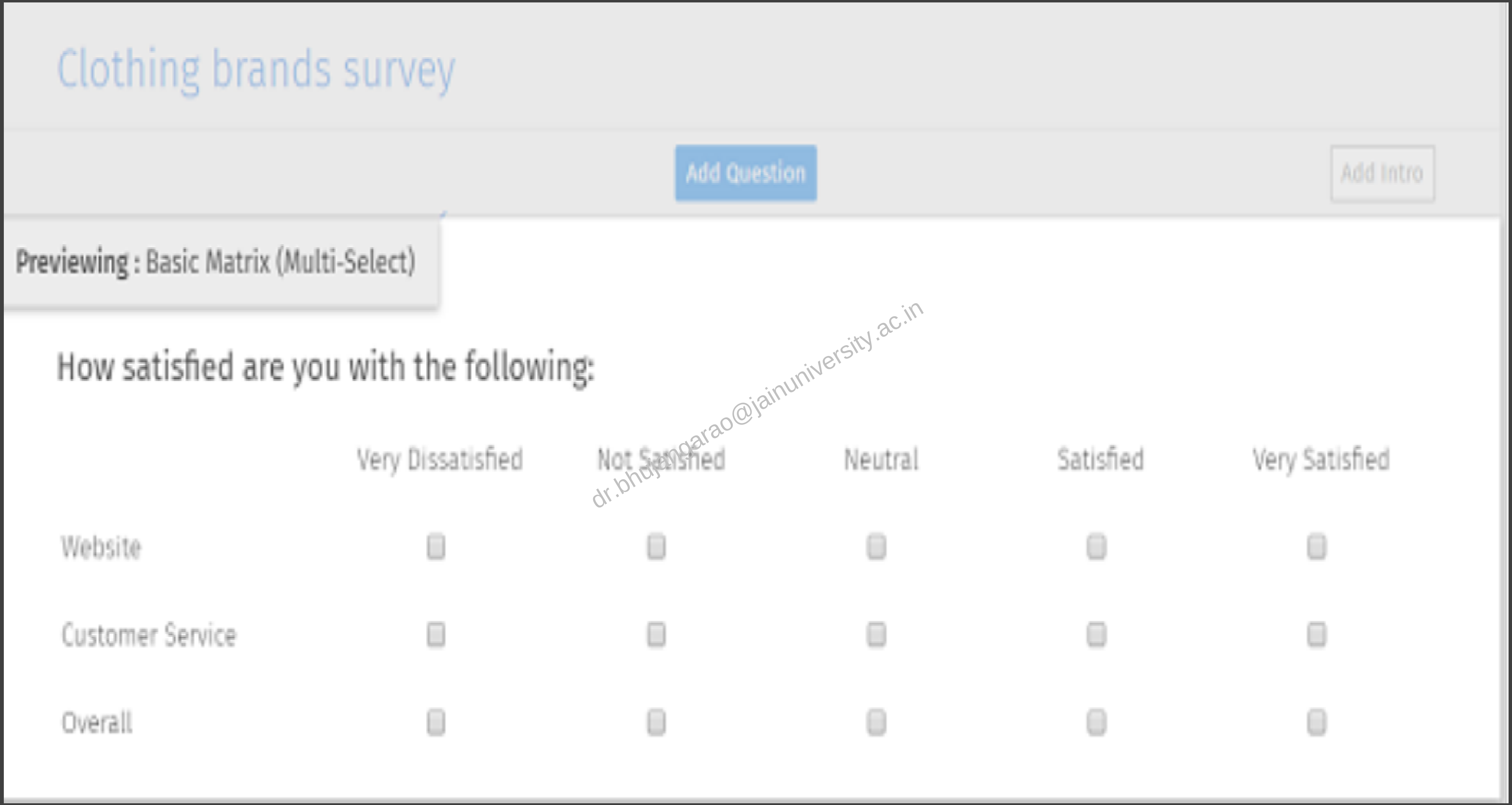
Quality

1. Very poor
2. Poor
3. Below average
4. Average
5. Above Average
6. Good
7. Excellent

Importance

1. Very unimportant
2. Unimportant
3. Slightly unimportant
4. Neutral
5. Slightly important
6. Important
7. Very important


NET PROMOTER SCORE
1
2
3
4
5
6
7
8
9
10
DETRACTORS
PASSIVES
PROMOTERS
NPS = % – %


Clothing brands survey
Add Question
Add Intro
Previewing : Basic Matrix (Multi-Select)
How satisfied are you with the following:
Very Dissatisfied
Not Satisfied
Neutral
Satisfied
Very Satisfied
Website
Customer Service
Overall

Ratio Scale:

The ratio scale is the 4th level of measurement scale, which is quantitative. It is a type of variable measurement scale. It allows researchers to compare the differences or intervals. The ratio scale has a unique feature. It possesses the character of the origin or zero points.

Characteristics of Ratio Scale:

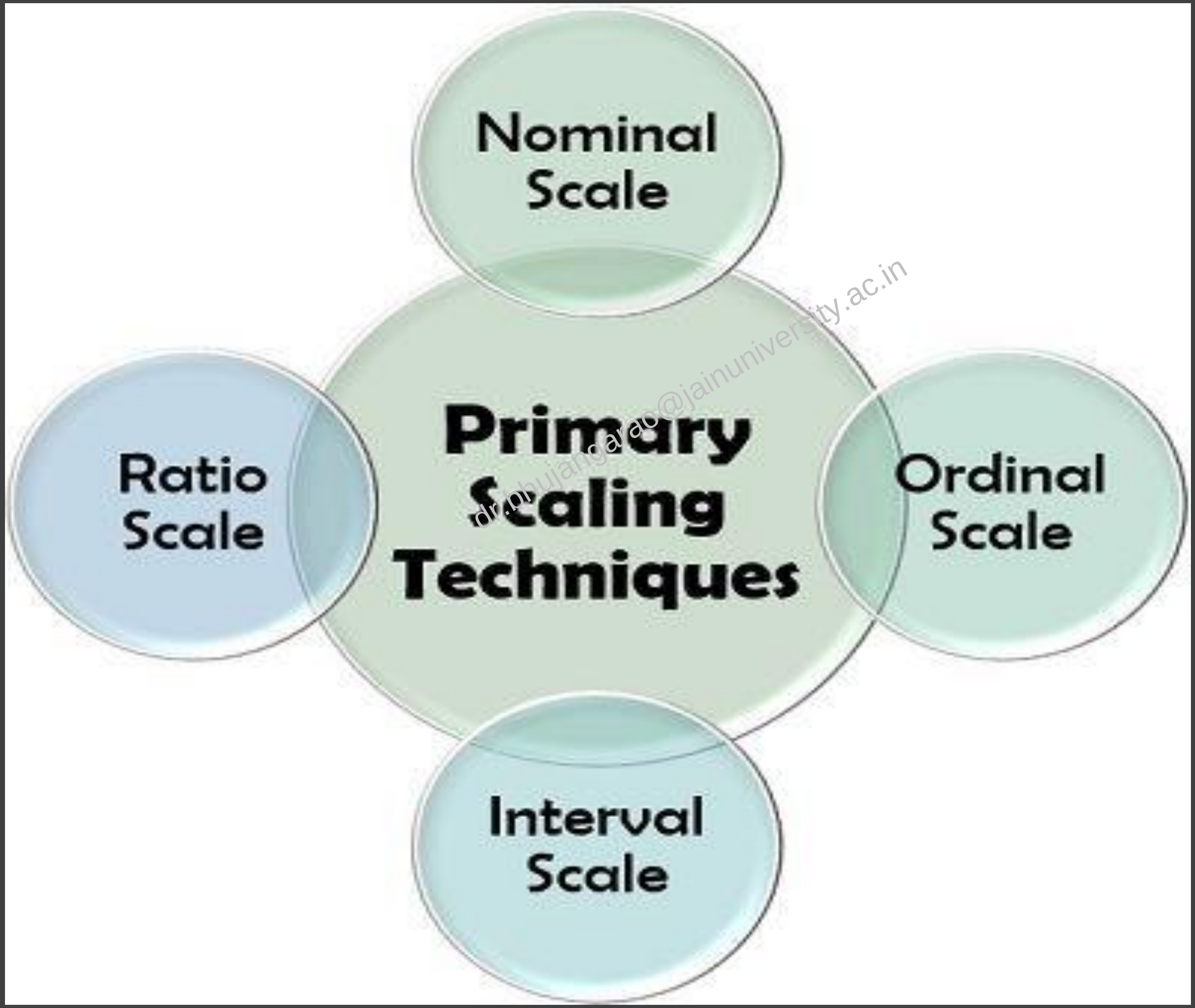
- Ratio scale has a feature of absolute zero
- It doesn't have negative numbers, because of its zero-point feature
- It affords unique opportunities for statistical analysis. The variables can be orderly added, subtracted, multiplied, divided. Mean, median, and mode can be calculated using the ratio scale.
- Ratio scale has unique and useful properties. One such feature is that it allows unit conversions like kilogram – calories, gram – calories, etc.

Example:

An example of a ratio scale is:

What is your weight in Kgs?
- Less than 55 kgs
- 55 – 75 kgs
- 76 – 85 kgs
- 86 – 95 kgs
- More than 95 kgs

Primary Scaling Techniques



NOMINAL SCALE

Nominal scales are adopted for non-quantitative (containing no numerical implication) labelling variables which are unique and different from one another.

Types of Nominal Scales:

1.Dichotomous: A nominal scale that has only two labels is called 'dichotomous'; *for example*, Yes/No.

2.Nominal with Order: The labels on a nominal scale arranged in an ascending or descending order is termed as 'nominal with order'; *for example*, Excellent, Good, Average, Poor, Worst.

3.Nominal without Order: Such nominal scale which has no sequence, is called 'nominal without order'; *for example*, Black, White.

Ordinal Scale

The ordinal scale functions on the concept of the relative position of the objects or labels based on the individual's choice or preference.

For example, At Amazon.in, every product has a customer review section where the buyers rate the listed product according to their buying experience, product features, quality, usage, etc.

The ratings so provided are as follows:

- 5 Star – Excellent
- 4 Star – Good
- 3 Star – Average

Interval Scale

An interval scale is also called a cardinal scale which is the numerical labelling with the same difference among the consecutive measurement units. With the help of this scaling technique, researchers can obtain a better comparison between the objects.

For example; A survey conducted by an automobile company to know the number of vehicles owned by the people living in a particular area who can be its prospective customers in future. It adopted the interval scaling technique for the purpose and provided the units as 1, 2, 3, 4, 5, 6 to select from.

In the scale mentioned above, every unit has the same difference, i.e., 1, whether it is between 2 and 3 or between 4 and 5.

RATIO SCALE

One of the most superior measurement technique is the ratio scale. Similar to an interval scale, a ratio scale is an abstract number system. It allows measurement at proper intervals, order, categorization and distance, with an added property of originating from a fixed zero point. Here, the comparison can be made in terms of the acquired ratio.

For example, A health product manufacturing company surveyed to identify the level of obesity in a particular locality. It released the following survey questionnaire:

Select a category to which your weight belongs to:
Less than 40 kilograms
•40-59 Kilograms
•60-79 Kilograms
•80-99 Kilograms
•100-119 Kilograms

Primary Scales of Measurement

Scale					
Nominal	Numbers Assigned to Runners				Finish

Ordinal	Rank Order of Winners				Finish

		Third place	Second place	First place
Interval	Performance Rating on a 0 to 10 Scale	8.2	9.1	9.6
Ratio	Time to Finish, in Seconds	15.2	14.1	13.4

A Classification of Scaling Techniques

- Scaling Techniques
  - Comparative Scales
    - Paired Comparison
    - Rank Order
    - Constant Sum
    - Q-Sort and Other Procedures
  - Non-comparative Scales
    - Continuous Rating Scales
    - Itemized Rating Scales
      - Likert
      - Semantic Differential
      - Stapel

PAIRED COMPARISION SCALE

Coke–Pepsi

Coke–Sprite

Coke–Limca

Pepsi–Sprite

Pepsi–Limca

Sprite–Limca

Brand	Coke	Pepsi	Sprite	Limca
Coke	—	ö		
Pepsi		—		
Sprite	ö	ö	—	
Limca	ö	ö	ö	—
No. of times preferred	2	3	1	0

RANK ORDER SCALE

2. Rank the following in order of preference

- ▼ Football
- ▼ Rugby
- ▼ Cricket
- ▼ Boxing

Constant Sum Scale:

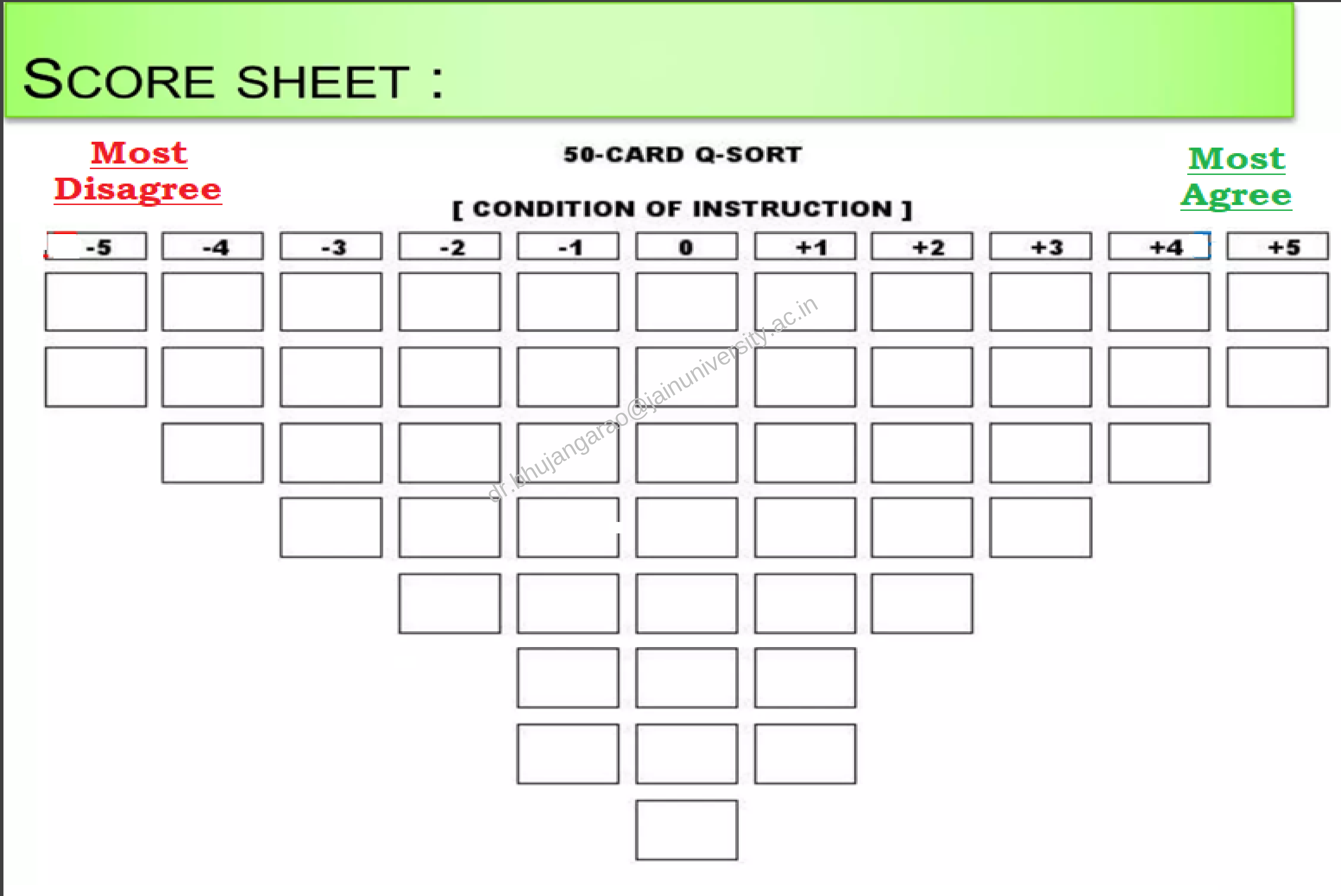
Importance of detergent attributes using a constant sum scale

Instructions: Between attributes of detergent please allocate 100 points among the attributes so that your allocation reflects the relative importance you attach to each attribute. The more points an attribute receives, the more important the attribute is. If an attribute is not at all important, assign it zero points. If an attribute is twice as important as some other attribute, it should receive twice as many points.

Format:

Attribute	**Number of Points**
(a) Price	50
(b) Fragrance	05
(c) Packaging	10
(d) Cleaning Power	30
(e) Lather	05
Total Points	100

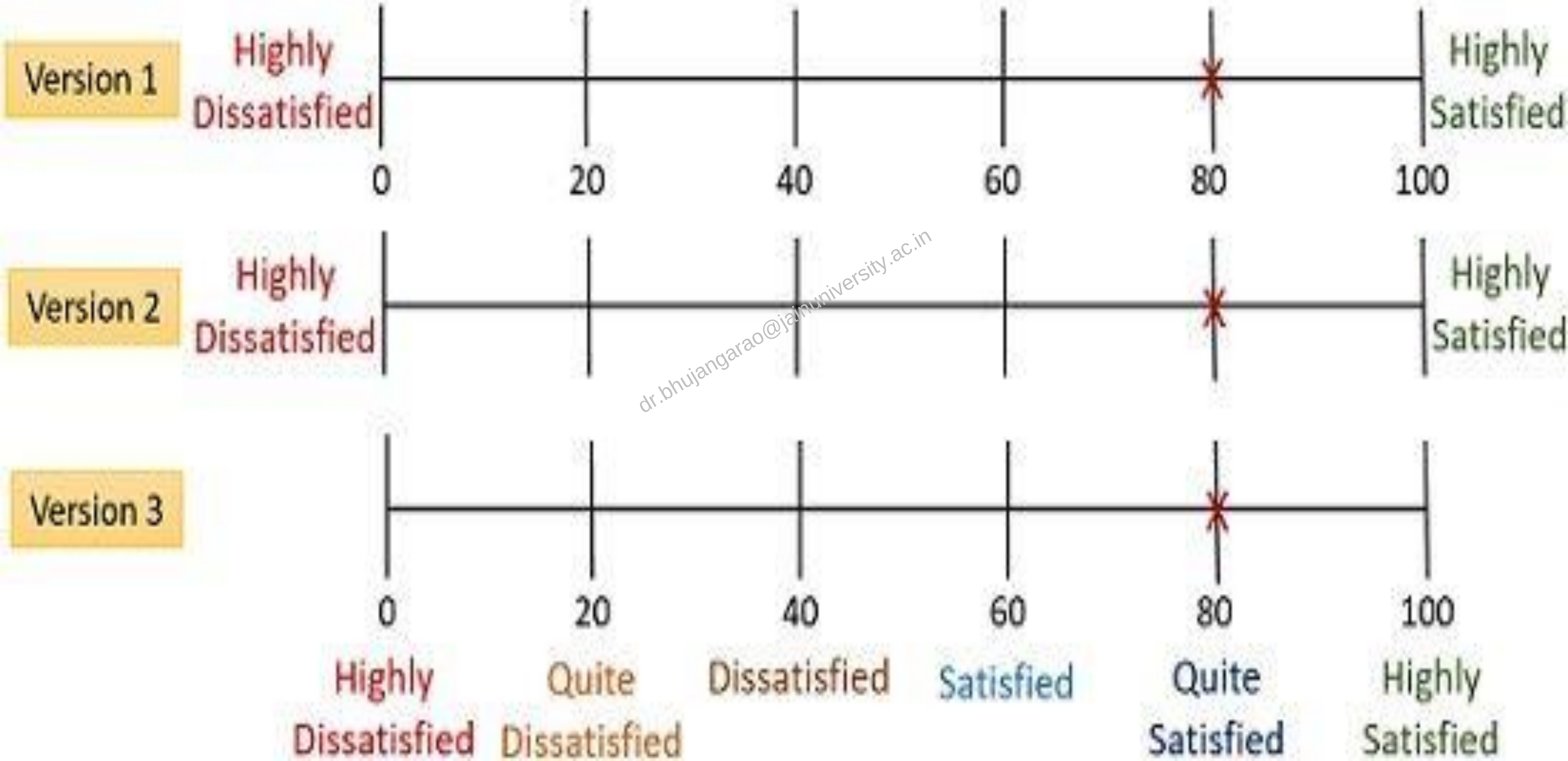
Q-SORT METHOD



DISAGREE AGREE

-4	-3	-2	-1	0	+1	+2	+3	+4
(26) The kidney is a black box	(23) The kidney makes sense	(24) Nothing is too difficult for me to learn	(30) Going to class is a waste of my time	(3) I am a visual learner	(21) By graduation, I should know how to interpret lab tests correctly	(28) I learn better when I'm with other students	(34) I wish other courses were taught like this one	(22) I will likely have to care for patients with kidney disease
(7) I am an auditory learner	(4) Our lectures are stimulating	(32) Our lectures are well organized	(10) Our instructors don't seem to enjoy teaching	(25) Physiology is easier than other courses	(12) Too much of what we learn is not clinically relevant	(20) I enjoy thinking about electrolyte problems	(9) I learn much of what is needed for class from textbooks	(5) I learn much of what is needed for class from attending the lectures
	(15) I would like to learn more about nephrology	(29) I learn in my own unique way	(2) I prefer to learn through reading or writing	(8) It will be important for me to understand the kidney	(14) Interactions during class help me understand concepts	(33) All of my questions get answered	(27) It is worth my time to come to class	
		(1) I am satisfied with how well I understand physiology	(31) The school supports students with different learning styles	(11) I have a good idea of what a nephrologist does	(36) After this course, I can forget about the kidney	(19) I do not understand acid-base disorders		
			(13) I learn much of what is needed for class from online lectures and videos	(16) I would be willing to shadow a nephrologist	(17) I am unfamiliar with the procedures a nephrologist performs			
			(37) I would like to hear about kidney disease research	(18) I learn much of what is needed on my own	(35) I have enough to time to keep up with everything			
				(33) All of my questions get answered				

Continuous Rating Scale



Likert Scale

5 Point Likert Scale

Numbering	1	2	3	4	5
	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
I like Stock Market	○	○	○	○	○
I like Stocks	○	○	○	○	○
I Like Money	○	○	○	○	○
I Like Return	○	○	○	○	○
Total	4	8	12	16	20

SEMANTIC DIFFERENTIAL SCALE

Democratic party

Bad __:__:__:__:__:__:_x_ Good

Cruel __:__:__:__:__:_x_:__ Kind

Unpleasant __:__:__:__:_x_:__:__ Pleasant

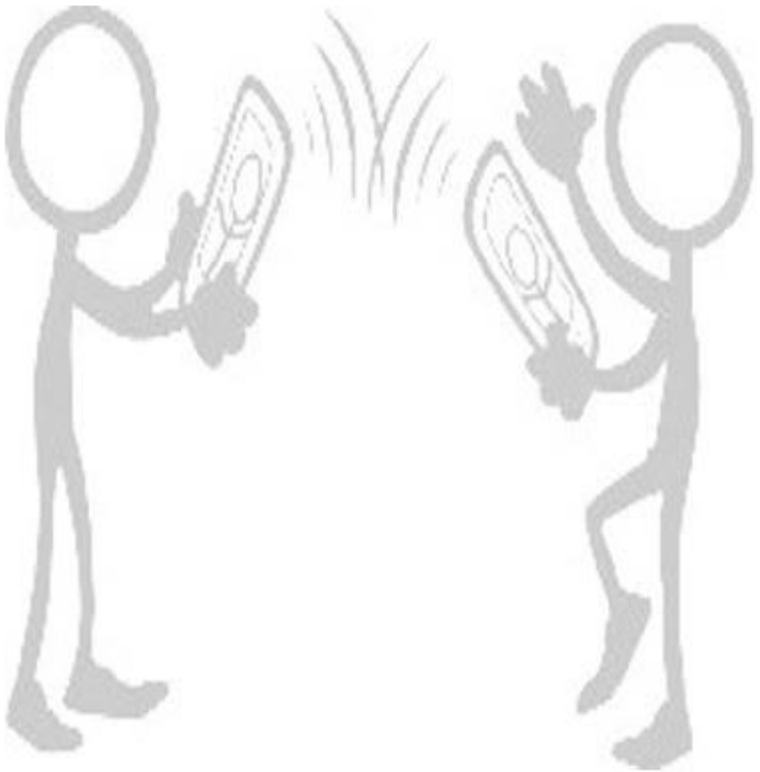
Unfair __:__:__:__:__:__:_x_ Fair

Dirty __:__:__:__:__:_x_:__ Clean

Negative __:__:__:__:_x_:__:__ Positive

Foolish __:__:__:__:__:__:_x_ Wise

1	2	3	4	5	6	7



STAPEL SCALE

Range from -5 to 5 with no 0.

Indicate how accurately or inaccurately each term describes our store

Cheap

-5 -4 -3 -2 -1 1 2 3 4 5

Quality Merchandise

-5 -4 -3 -2 -1 1 2 3 4 5